

IKT systemy - Windows

Úvod virtualizácie

Hypervisor = softer code which is used for bare metal virtualization

vHost = virtualization host, physical computer with installed hypervisor

VM = virtual machine, is logical system with own guest OS running on vHost

Uplink = physical network adapter attached to vHost

Datastore = storage device used to store VMs, ISOS,..

vCPU = virtual CPU

vRAM = virtual RAM device of VM

vDisk = virtual HDD device of VM

vNIC = virtual NIC device of VM

vSwitch = virtual switch used by vNIC (left side) and Uplinks (right side)

Appliance = closed image (VM) with pre-installed guest OS and application

Virtualisation is technology, which allows to run multiple logical instances with own OS and Applications installed (VMs) on single or clustered physical.

Pros:

- System consolidation
- Hardware independent software
- High scalability and availability
- Better usage of hardware resources
- Lower costs of hosting, cooling, power supply
- Software based datacenter with central management
- Snapshots (backup)

Cons:

- Higher investment
- Not all physical components can be connected to VM (special hardware no)
- Not all application can run on virtual platform

Virtualisation types - Bare metal (Type1) → VM a hypervisor - Guest OS based
(Type 2) → on ma virtualization SW a
application az potom OP

Softer define datacenter - softverove riesenie

Virtualisation - ESXi

System Requirements (min)

- 64 bit CPU dual core
- Intel or AMD Virtualisation tech. Enabled
- Data Execution Prevention (DEP), NX/XD bits enabled in BIOS
- 4 GD of Memory

NIC (1GB, 10GB, 40GB are supported) —> sietova karta

SLA —> pre zakaznika

OLA —> pre nas interne —> 1. Dostupnost, 2. Resolution time, 3. Reaction time

2 types: **STANDARD**

Needs to be configured per vHost

Not required VCS server for configuration

DISTRIBUTED

Required VCS server for configuration

Shared configuration across vHostgso

STORAGE VMware ESXi

—> **ESXi Hosts**

—> **Datastore Types** (VMFS, NFS -> File systems)

—> **Storage Technologies** (Direct attached, Fibre Channel -> optika, FCoE -> nieco medzi, iSCSI -> data posiekane po sieti , NAS-> storage box)

VM Files VMware ESXi

Configuration file .vmx / pri virtualke .vcm

Disk files .vmdk + -flat.vmdk .vhdx

Bios file .nvram

Swap file .vswp

Snapshot files .vmsd (data) + .vmsn (state) + -delta.vmdk

Log file .log

Installing and configuring: Hyper-V

Hyper-Vserver

Standalone product

is available for free

Licensing Windows server

Standar 2 OSEs

Datacenter unlimited OSEs

Requirements vHost

Hardware (min)

64bit CPU with SLAT and support virtualisation tech

DEP enabled (XD or XN tech)

4 GB RAM

100 GB HDD

Software

Installed Windows server 2016 w Hyper-V feature or Hyper-V Server 2026

Major/Core feature

Production checkpoint

Default option
Point in time image

Standard Checkpoint

Chose capturing SAVE state of VM
Usable for test, development, resistant APP

Supported TECH

1. iSCSI
2. FC
3. SMB 3.0 shares
4. Shared VHDX

Supported partition format

1. NTF
2. REFS

CSV musime vytvarat

Software based layer 2 ethernet network switch

Traffic connectivity

EXTERNAL

All communication outside

INTERNAL

Allow communication with VMs connection

PRIVATE

Same switch

Limitations

Needs to have at least one physical adapter
You cannot attache pNIC to multiple Switches
You can use VLAN taking

AVHDX

different

VMCX

Naming convention - pomenovanie podľa toho na čo slúži
WS rc CPU - 2 → Virtual WS CPU - 4

OVERPROVISIONING - 1GB:2 predávam niečo čo nemám \$ → kapacitný monitoring

Disk provisioning - Thin provision → postupne naplná storage, keď vymažeme vráti sa nám miesto

Thick provision → rovno zaberie celý hard disk prázdny miestom a keď vymažeme miesto sa nevráti

Ake typy workloadov môže byť na hyperv?

Konfiguracné files ďalší disk, zálohovanie, rýchlosť

Enhanced session policy policy -
Replication configuration - asynchrónna

Generations

1. 32-bit pre staré systémy
2. 64-bit najnovšie generácie

Dynamic memory - využíva RAM naplno, pridáva sa mu RAM ako potrebuje a zvyšok pridá inej AP = môžeme pridať viac VM 8G ako máme RAM 4G

AD DS - OVERVIEW

What are AD DS? - Active Directory, domain services

Why do company needs AD DS services - what are the benefits? Pre konfiguráciu

When do you need more than one domain? viaceré dcérskych spoločností, test and production

Exist some relation with DNS? - domain naming, service location records, DCD, replication traffic, namespace consistency,

Forest vs Domain - spojenie viacerých domén - obsahuje v sebe kľúčové kontroly 2

Administrators of forest/domain - práva všade

Database and replication - v rámci domény

AD components

Physical components

- **DC** | contains read-write copy of AD database
- **RODC** | special instance of DC with read-only database copy
- **Global catalogue** | special DC with host GC role which contains read

copy all objects in forests for speed up searching between domain

- **Datastores** | holds AD database Ntds.dit and log files by default on all DC in C:\Windows\NTDS + Sysvol

Logical components

- **Domain** | security boundaries for users, computers, other objects
- **Domain tress** | collection of domains witch shares same root domain name and DNS namespaces
- **Site** | collection of object, defined by their physical location
- **Forest** | collection of domains witch share common AD DS
- **OU** | administrative object for group obj, delegate permission and applying GPOs
- **Containers** | same like OU, but it cannot be applying GPOs

Domena - **domain admins** -> prava na domene

Forest - **enterprise admin** -> maju prava vsade

Post-dep-config

- Hostname
- DNS suffix
- IP (static)
- FS layout
- Local Users
- Softverovy firewalll
- Activation OS
- Domain Join
- Set TimeZone
- Set time

slmgr -xpr -> na zistenie ci ze aktivovany system

slmgr -ato -> na aktivaciu systemu

cd \. set-NetfirewallProfile - Enabled False -All

slmgr - ato

slmgr -rearm

Result set of policy -

192.168.118.18 -> moja IP

DNS

- 3 domains (1 je root)
- Minimalne 2 musia nainstalovat (s redundanciou 4)
- 2016 verzia priniesla nove ficurky - 2012 je stara verzia kde su vsetko domeny na jednej urovni a nie su tam fickurky
- Aspon 1 global controller

Sposob obnovenia domain controller

Aut - DC replikuje na existujuce -> ked chceme aby jeden prepisal ostatne

Nout - DC si zreplikuje z existujucich na seba -> pouzivame ak nam jeden vypadne, realne sa nepouziva

NETBIOS NAME zoberie prvych 15 znakov

Pri instalacii **promo controller** je potrebný restart

Na domain controllers sa nemozeme prihlasit ako normalny uzivatel ale ako admin

V každej domene musi byt DNS a DC - koli redundancii 4 (2 z kazdeho)

Aspon 2 foresty

VC.DUAL.SK-KOSICE.T-SYSTEMS.SK.

VC. - hostname

DUAL. - DNS

SK-KOSICE.

T-SYSTEMS.

SK. -

. - root

Inter replikacia - dnuka

Intra replikacie - vonku

Geograficka poloha

Workload - test production

Administrativa - aby sme to nespravovali sami

Multi master mode - DC funguju na <- z neho moze kontrolovat vsetky DC vo foreste —> musime mat zabezpecenu vysoku rychlost

DSRM - f8 tlacitko -> zadat heslo -> cez comand

Aky prikaz na restor potrebujeme? —> ntdsutil

Aky je rozdiel medzi kontainerom a OU

Prehľadnosť, administrácia a (aplikovat group polici ale iba na OU)

GP umoznuju nastavovat pasword polici, account polici , instalovat soubor .msi

Domain and trust - na prepojenie dvoch domain trees—> ak chceme suport od

niekoho my ideme ku nemu takže zadávame cestu → k nemu.

Domain Site and servicer - define SUB NET cez DC a hľadá ho najprv u seba a potom inde cez IP adresu

Active directory administrativ center - nahrádza čiastočne základné úkony cez GUI

→ enable recycle bin

Users and computers - práca s objektami - kontajnery, OU

TGT - vytvára sa pri prihlasovaní a má tam informácie o právach, skupine atď.

SSO - single site on → vieme sa prihlásiť cez mail

Priradovanie viacej práv v doméne US sú kumulatívne - na kontrolu všetkých sa každý dáva do USERS

UG → pridelenie oprávnení
→ logické zostupenie

OU - organisation unit

Powershell

Zmeniť meno a IP adresu - DNS konci 10 a predtým 100 podľa protokola

C:\Windows\System32\....?

Group policy - lokal GP - computers settings, user settings
- domain GP

Gpedit.msc

Windows settings - scripts - startup

- **shutdown** → počas oboch môžeme dať

nejaký skript

- **Security settings - account policies** - account

lockout policy

- **Password policy** - koľko písmen atď.

Některé politiky sa urobia po odhlásení alebo po nejakom case (90 min) heslo, niektoré až po reboot

GROUPUPDATE | FORCE - prejaví sa hneď bez reboot

GP result - môžem si to vypracovať cez HTML

Linkovanie novej policy mozeme na uzadi? - child OU

- site
- domain
- OU

Groupupdate - aby som updatoval GP

Ked mam viac policy ta mensia, spodnejsia vyhrava

Get-command —> vypise vsetky mozne prikazy

POWERSHELL

Objektovo orientovane

Zavisly .NET

Not case sensitive

SYNTAX

Verb - Noun —> co? - s cim?

Parameter <value>

Get command - verb - noun

VERB

|> co chceme urobit?

NOUN

|> hladat cim?

1. Get-localUser
2. Get-Volume
3. Get-Service
4. Get-Member - vypise vsetky mozne prikazy

Start-Transcript - Path C:\Temp\PowerShell-Day2.txt —> zaznamenava logy co sa pise

Get-command Get Service —> Get-Service

Ako pracovat s PS

1. Najst chdlet
2. HELP —> Get-Help (-ShowWindow) -Online

Get-service -name wuauserv , RpcEp...

Get-Host —> vypise verziu PS

WhatIf —> vypise co by sa stalo ak by sme pouzili konkretny command

Write-Host "I love PoweeShell with Dual members of APP group "

-ForegroundColor Magenta

New-Variable —>

PREMENNE

\$<NazovPremennej>

\$Meno = "Stanislav"

\$Priezvisko = "Lukac"

&Vek = 18

\$PSVersionTable

- \$Meno.GetType() —> na zistenie aky je to string

-\$Priezvisko | Get-Member —> ukaze metody

-Int32

ZADEFINOVANIE KODU CIM BUDE

[int]\$Vek = \$null

[string]\$Name = \$null

PSP

-VAR

-ALIAS

Praca s objektami

command#1 | command#2

Musi mat vystup | musi vediet pracovat s vystupom

PLAN A - by value, musi dokazat s nimi pracovat

PLAN B - by property name

Get-service -name wuauserv | stop-service -whatif

FILTROVANIE:

1. Krok —> Get-

2. Krok —> filtrovanie | select-object, where-object

3. Krok —> sort - object

4. Krok —> out | export

Bud jedno alebo druhe !

1. Krok —> format | format-list, format-table

Select-object —> dokaze nam vyfiltrovat prvych alebo poslednych Get-

service | select-obj -last 2 | select obj - property -name, displayname | gm

Where-object —> vyfiltruje podla presnych parametrov... Get-Process | Where-

Object {\$_.CPU -gt 100}

Where-object—> na zaklade —> get-service | gm -membertype property

Get-service | where object {\$_starttype} —eq "automatic" }) .count | where-

object {\$_status -ne "running"}

\$_ -> plati na vsetky vystupy z lavej strany

Get-childitem -path C: | sort-object -prop length | select-object -first 5

Export-clicxml —> dokaze pracovat s objektem, dokaze pracovat s polami

Export-Csv —> na ulozenie aby sme s tým potom vedeli dalej pracovat, zachovava member

Get-content —> nacitanie kontentu.. potom sa to meni na .txt (string), nedokaze nic

PowerShell

SLUCKA FOR EACH

#foreach-object

#syntax: foreach (\$obj in \$objcoll) {<scriptCODE>}

```
$services = Get-Service
```

```
Foreach ($obj in $services) {  
    write-hosts "services $obj.Name) has status $($obj.Status)"  
}
```

#for

#syntax: for (\$init; condition; \$Ops_\$I) { <code> }

```
For ($I = 0; $I -lt 5; $i++) {  
    "[I] Services $($services[$I].Name) has status $services[$I].Status"  
}
```

#dowhile

```
$services = Get-Service
```

```
$i=0
```

```
Do
```

```
{
```

```
    write-hosts "[I] Services $($services[$I].Name) has status  
$services[$I].Status"
```

```
    $i++
```

```
}
```

```
while ($i -lt 5)
```

#dountil

```
Do
```

```
{
```

```
    write-hosts "[I] Services $($services[$I].Name) has status  
$services[$I].Status"
```

```
    $i++
```

```
}
```

```
until ($i -gt 4 )
```

#while

```
while ($x -lt 1)
{
    Write-Host "I hate PowerShell" -ForegroundColor Red
}
```

#Podmienovaci vyraz

```
If ($x -eq 99) {
    Write-Host "[ $l % complete] Analysing system, ... please wait"
    $l++
}
elseif ( ) {
    codeB
}
else {
    codeC
}
```

SWITCH (R-premenna) {

```
X1 {codeA}
X2 {codeB}
Default {code3}

}
```

PS -layout

1. Capacity [GB]
2. Free space [%]
3. Free space [GB]

Get-volume | where-object {\$_.Driveletter -ne \$null} | select-object - property driveletter, Size*

1. Get-volume | where-object {\$_.Driveletter -ne \$null} | select-object - property driveletter , @{name="Capacity[GB]"; e={math::(\$_.Size/ 1Gb,2)}
2. Get-volume | where-object {\$_.Driveletter -ne \$null} | select-object - property driveletter , @{name="Freespace[%]"; e={math:: (((\$_.SizeRemaining/\$_.Size)*100,2)}}}
3. Get-volume | where-object {\$_.Driveletter -ne \$null} | select-object -

property driveletter , @{name="Freespace[GB]"; e={[[math]:: (\$_.SizeRemaining/1Gb,2)}}}

New Recording

27 Nov 2024 at 9:22

Audio · 0s



CIM - celosvetovy standart → pristup k info o hardvery alebo softvery → powershell remouting 2 TCP porty
Port → 5985
Ak ideme z vonku → 5986
CIM-instance

New Recording 2

27 Nov 2024 at 10:16

Audio · 0s



One too many - invoke command →
MTP session -

2025 - windows server

VPN

Protocol - open VPN, ipsek (balik bezpecnostnych protocolov), autetifikacne protocoli,

Open VPN - Komunikacia s VPN serverom a VPN klientom, cez verejnu siet internetu, privatne pripojenie

PAP - pre developera ak skusaju nieco co nie je citlive, NEKONTROLUJE NAZACIATKU INFO O TOM KTO POSIELA

Typy VPN

Site to site - prepojenie 2 a viacerych site

Point to site - pripojenie jednotlivcov do situ, vasinou nemaju dedikovane miesto

Ake komponenty potrebujeme na POINT TO SITE

- 1.VPN server —>
- 2.VPN klient —-> inclu VPN tunel
- 3.DC
- 4.DHCP server (VPN samostatne moze mat dhcp v sebe a vie pridelovat ip adresy)

VPN server -> na neho sa pripaja klient cez internet

NPS - validacia ci ten kto sa pripaja ma up to date patche, ak nie prideli mu ip na, ktorej si doinstaluje patche a znova sa klient pripaja na NPS kde ho znova skontroluje a az po overeni ho pusta dalej do siete

DC

DHCP

WINDOWS SERVER STORAGE DOCUMENTATION

<https://learn.microsoft.com/en-us/windows-server/storage/storage>

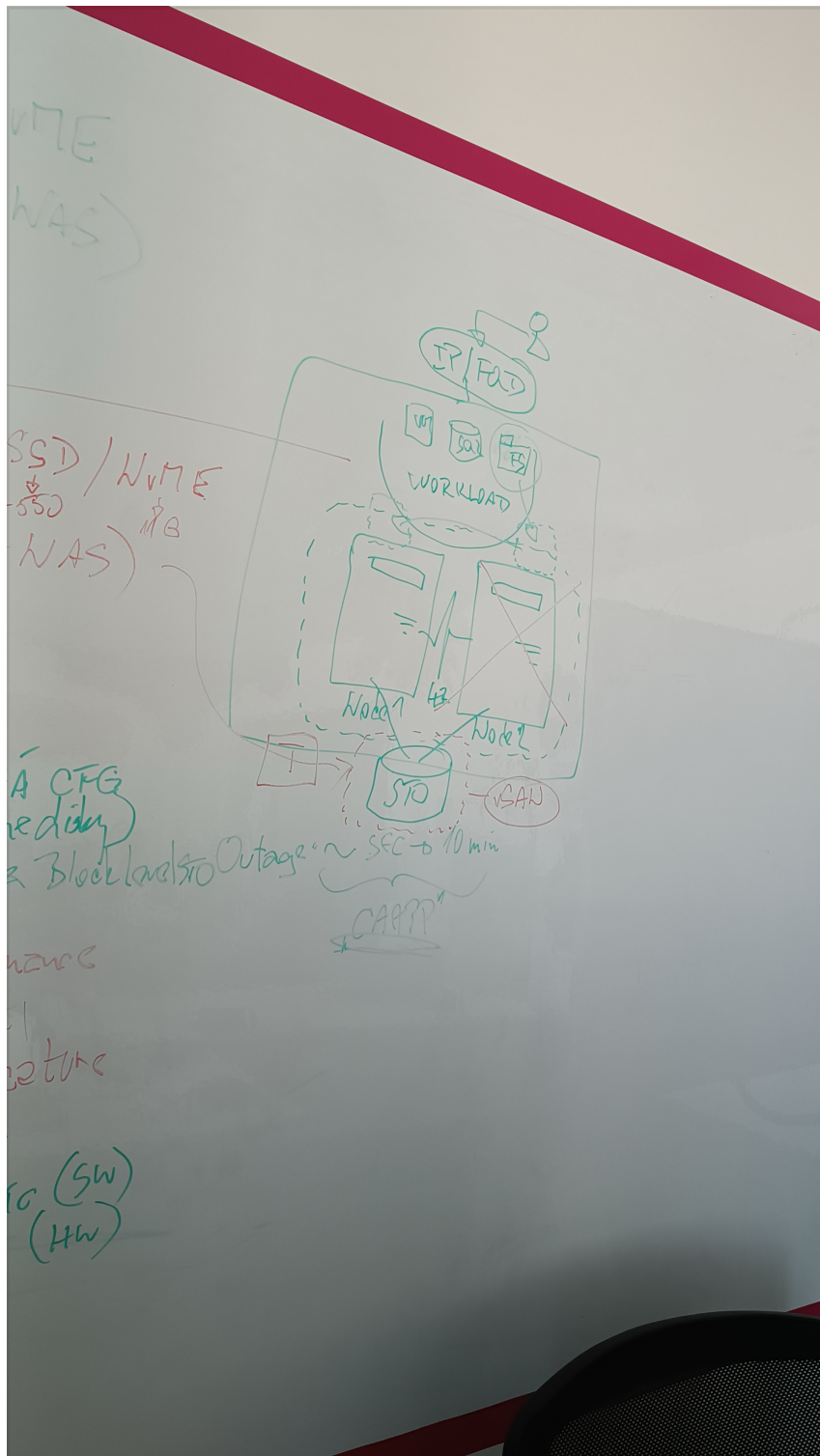
Domace riesenia

- HDD / SSD / NVME
- Ext. NAS, synology
- Cloud

Produkcia

- SAS (HDD) 15000 otacok, 24/7 / SSD / NVME
- Ext. SAN+NAS
- Cloud

Closet avare application



ISCSI

- ako keby nahradza 2. riesenie

VYHODY A NEVYHODY

1. Jednoduchá konfigurácia.
2. cena lokálneho disku
3. Licence free, súcast windows servera and block level
 - Rychlost / performance

- Kapacita
- Enterprise feature

REQUIRMENTS

- Windows srv licencia
- Disk na, ktore sa budu ukladat data
- TCP IP - konekcia pre disky

I:/ volume —> raid 1 pre redundancy

STORAGE SPACES

Storage pool —> disky su sucastou jedneho poolu kde mozu byt raidy 0/1/5. —> to cele sa pripaja ako sietovy disk



DFS - X → DFS - N (centralne spravovanie), SMB shares Access Point

DFS - N → logic groups kde je UNC Parth pre uzivatela. \\ 1 IP/FQDN \ 1 nameSMD

DFS - R → replikuje SNB shares

1. HA without clu
2. Btw branch-offices
3. Central replication to sites
4. Easy management

Klaster

Stav

active/active

active/passiv

Requirements :

Preco chceme mat 2 klastrovane serveri. Nod A/ nod B

Failover klaster

1. No single point of failure
2. Utilisation
3. Aby sme vedeli kde je kotra aplikacia

Witness -> LUN from storage box (1GB)

SMB share

Azure storage account

Len aby mal 1 hlas , pri pade 1 klastra aby bol stav 2/3 a nie 1/2, vzdy je iba jeden ale more byt zdielany medzi klastre.

NLB klaster (network load balancer)

pre webove aplikacie, autefikacne servre

Potrebuje rozlozit load aby nespadli servere od zatazenia od pouzivatelov

Max 32